It states that "If a system at equilibrium is disturbed, then the system shifts itself in a direction that nullifies the effect of that disturbance."

8.8. 1 Effect of concentration

At equilibrium, the concentration of the reactants and the products does not change. The addition of more reactants or products to the reacting system at equilibrium causes an increase in their respective concentrations.

According to Le Chatelier's principle, the effect of increase in concentration of a substance is to shift the equilibrium in a direction that consumes the added substance.

Let us consider the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The addition of H_2 or I_2 to the equilibrium mixture, disturbs the equilibrium. In order to minimize the stress, the system shifts the reaction in a direction where $H_2$and I_2 are consumed. i.e., the formation of additional HI would balance the effect of added reactant. Hence, the equilibrium shifts to the right (forward direction) i.e. the forward reaction takes place until the equilibrium is re-established. Similarly, removal of HI (product) also favours the forward reaction.

If HI is added to the equilibrium mixture, the concentration HI is increased, and system proceeds in the reverse direction to nullify the effect of increase in concentration of HI .

Let us explain the effect change in concentration by considering the formation of HI from H_2 and I_2. At equilibrium, the concentrations of HI, H_2 and I_2 are 1 M, 0.2 M and 0.1 M respectively.

$$K_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{1 \times 1}{0.2 \times 0.1} = 50$$

The equilibrium was disturbed by adding 0.1 M iodine to the reaction mixture. After sometime, the concentration of HI is found to be 1.092 M. Let us verify whether the system proceeds to re-establish the equilibrium according to Le Chatelier's principle.

	H_2	I_2	HI
Concentration at equilibrium	0.2	0.1	1
Disturbance (Addition of 0.1 M iodine)		+0.1	
Concentration immediately after the addition of iodine	0.2	0.2	1
Change in concentration	-x	-x	2x
New concentration	0.2-x	0.2-x	1+2x

Now,

Concentration of HI = 1+2x = 1.092 M

2x = 0.092

x = 0.046 M

Therefore, the concentration of hydrogen and iodine at this stage,

$[H_2] = 0.2 - X = 0.2 - 0.046 = 0.154$

$[I_2] = 0.2 - X = 0.2 - 0.046 = 0.154$

11th Chemistry Chapter-8 Vol-2 EM.indd 16 09-01-2023 14:38:27